



LESSON 14.4

COMPARING THEORETICAL PROBABILITY TO EXPERIMENTAL PROBABILITY

LESSON INTRODUCTION

MA.8.DP.2.1 Determine the sample space for a repeated experiment.

MA.8.DP.2.2 Find the theoretical probability of an event related to a repeated experiment.

LEARNING TARGETS

- I can calculate theoretical probability and compare it to experimental probability to evaluate accuracy.

BENCHMARK COHERENCE

BUILDING ON

MA.7.DP.2.1 Determine the sample space for a simple experiment.

MA.7.DP.2.2 Given the probability of a chance event, interpret the likelihood of it occurring. Compare the probabilities of chance events.

WORKING TOWARD

MA.912.DP.4.1 Describe events as subsets of a sample space using characteristics, or categories, of the outcomes; or as unions, intersections or complements of other events.

MA.912.DP.4.3 Calculate the conditional probability of two events and interpret the result in terms of its context.

ESSENTIAL QUESTIONS

- How does theoretical probability compare to experimental probability?
- How can you calculate the theoretical probability of multi-step experiments?

LESSON NARRATIVE

This lesson helps students form a concrete idea of how to calculate the theoretical probability of multi-step experiments. Students use these calculations to compare theoretical probability to experimental probability. Although they are not using experimental probability to evaluate accuracy in this lesson, an introduction to comparing the probabilities will provide a good understanding of evaluating the accuracy of theoretical probabilities calculated in successive lessons.

COMPONENT SUMMARY

14.4.1 WARM-UP (~5 MIN.)

Career Video – Using Statistics in the real world

14.4.3 COLLABORATION (10 MIN.)

Notice trends in calculating compound probability

14.4.5 WRAP-UP (~5 MIN.)

Write a note to summarize the lesson

14.4.2 EXPLORATION (20 MIN.)

Compare theoretical probability to experimental probability after completing an experiment

14.4.4 BRING IT TOGETHER! (5-10 MIN.)

Complete summary to make generalizations about how to calculate theoretical probability of multi-step experiments

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Experimental probability

MATERIALS

- Career Profile Video
- (Optional) Online spinner and coin flip application

ADDITIONAL PREPARATION

- 14.4.1 Warm-Up contains a Career Profile Video to accompany the question prompts.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may misinterpret or misuse theoretical probability for experimental probability.
- Students may not understand why the experimental probability is not the same as the theoretical probability.

THINGS TO CONSIDER

- In 14.4.3 Collaboration, the probabilities that are referenced are the same as the ones calculated in the 14.4.2 Exploration.
- In successive lessons, students will be comparing theoretical and experimental probabilities more, so provide them with the opportunity to not only compare the values, but also the methods for calculating them and the definitions.

LITERACY AND LANGUAGE ROUTINES

Compare and Connect

In the 14.4.2 Exploration, allow students to explore the comparisons of theoretical and experimental probabilities, as well as their calculated experimental probabilities, to that of another group. Prompt students to use mathematical language when having these discussions. This language routine supports meta-cognitive and meta-linguistic awareness and also supports mathematical conversation.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.5.1 Patterns and Structure

In 14.4.3 Collaboration, students bridge together what they have noticed in previous questions. Students look back on their previous answers to identify a pattern that leads them to develop a method for calculating theoretical probability for multi-step experiments. They will look for similarities among the questions and create a procedure, gaining important practice in this Math Thinking and Reasoning Standard.

DIFFERENTIATE INSTRUCTION

For ELL's, consider pre-teaching experimental probability. If students are given instruction about experimental probability ahead of time, along with a review about how to calculate theoretical probability, this can help them be more successful in the lesson.

Additional differentiation opportunities are denoted throughout the Lesson Guide.

14.4.1 WARM-UP: DATA ANALYST



14.4.1 Warm-Up: Data Analyst

Ivana Seric has her PhD in mathematics. She works as a data analyst for the Philadelphia 76ers, an NBA basketball team. Data science has changed the game of basketball as players and coaches have been able to use data and analytics, rather than instincts and talent alone, to determine how to improve the game. Data analytics is a growing career field in a world where more and more data are collected and shared through increasing technological means.



1. What are some ways data analytics has changed the game of basketball?

Answers will vary. Data analytics help coaches create different strategies to win games. Data analytics allows teams to better plan for opposing players as they can use them to determine players' or teams' strengths and weaknesses.

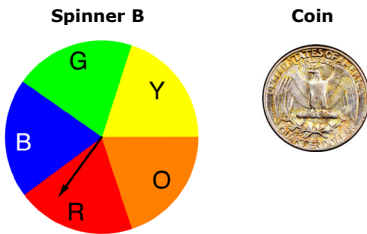
2. Explain how other industries might use statistics.

Answers will vary. Other industries may keep statistics on how many people use their products and the demographics of those populations. They may keep statistics on what day and time most people watch TV. They may use data analytics to determine which advertisements are most effective in selling their product or to determine the efficiency of employees.



14.4.2 Exploration: Calculating Theoretical Probability

Work with your partner to complete the following simulation involving spinning Spinner B and flipping a fair coin.



As part of a game, Spinner B is spun and a coin is flipped. The theoretical probabilities of events in this random process can be found by determining the sample space using a list, table, or tree diagram.

14.4.2 EXPLORATION: CALCULATING THEORETICAL PROBABILITY

TEACHER MOVES

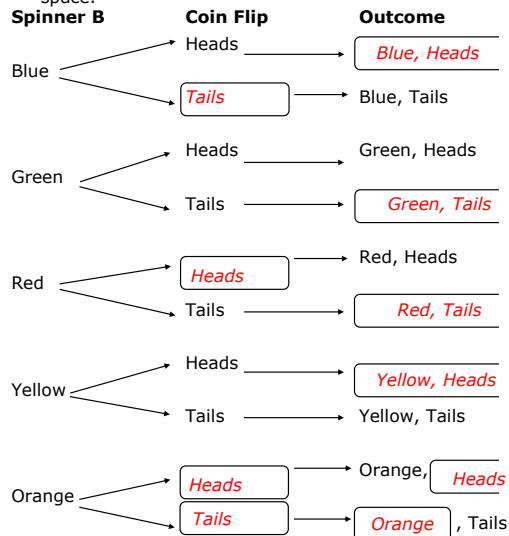
This Exploration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered later in the lesson. Students may revise their thinking throughout the exploration as they engage with each new reflection. Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole group. Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed Exploration.

ENRICHMENT

Help students who may struggle by asking scaffolding questions like:

- Before determining the sample space, how many outcomes will there be?
- How can you determine the number of outcomes?

1. Complete the tree diagram to determine the sample space.



2. Find the theoretical probability that the spinner lands on yellow and heads is flipped on the coin.

$$P(\text{yellow, heads}) =$$

$\frac{\text{the number of outcomes that are (yellow, heads)}}{\text{total number of outcomes in the tree diagram}}$

$$= \frac{1}{10} = 0.100 = 10\%$$

14.4.2 EXPLORATION: CALCULATING THEORETICAL PROBABILITY (CONTINUED)

ENRICHMENT

If you notice students struggling to work through this section, prompt student thinking with questions like:

- Which outcomes are included in calculating $P(\text{not blue, tails})$?
- Which outcomes are included in calculating $P(\text{primary color, tails})$?

QUESTIONING

Consider challenging critical thinkers with the following questions:

- Find three different events with the theoretical probability of $\frac{2}{5}$.
- What is more likely, $P(\text{not blue, tails})$ or $P(\text{primary color, tails})$?

DIFFERENTIATE INSTRUCTION

If technology is a limitation for the simulation, use scraps of paper with each outcome labeled. Students can draw, with replacement, to mimic the spinner.

- Find the theoretical probability that the spinner *does not* land on blue and the coin lands tails up, $P(\text{not blue, tails})$.

$$\frac{2}{5}$$

- What is the theoretical probability that the spinner lands on a primary color (red, blue, or yellow) and the coin lands tails up, $P(\text{primary color, tails})$?

$$\frac{3}{10}$$

One way to evaluate the accuracy of theoretical probabilities is to compare them to experimental probabilities.

- Work with your partner to conduct 50 trials of the game where Spinner B is spun and a fair coin is flipped. Using a fair decision-making process, decide which partner will be in charge of spinning the spinner and which will be in charge of flipping the coin. Record your data in the table by tallying each outcome in the middle column. After you have completed the simulation 50 times, record the total number of occurrences for each outcome. *Answers will vary.*

Outcome	Number of Times the Outcome Occurs	Experimental Probability of Each Outcome
Red, heads	Total: 6	0.12
Red, tails	Total: 7	0.14
Orange, heads	Total: 3	0.06
Orange, tails	Total: 4	0.08
Yellow, heads	Total: 7	0.14
Yellow, tails	Total: 5	0.10
Green, heads	Total: 3	0.06
Green, tails	Total: 6	0.12
Blue, heads	Total: 5	0.10
Blue, tails	Total: 4	0.08
Overall Total	50	1 = 100%

6. Find the experimental probability of each outcome in the sample space and record it in the last column in the table. For example, to find the experimental probability of the outcome "yellow, heads," divide the number of times the outcome occurred by the total number of trials. Suppose "yellow, heads" occurred 7 times in the experiment.

$$P(\text{yellow, heads}) = \frac{7}{50} = \frac{\boxed{14}}{100} = \boxed{.14} = 14\%$$

7. Find the experimental probability that the spinner *does not* land on blue and the coin lands tails up, $P(\text{not blue, tails})$.

Answers will vary.

$$P(\text{not blue, tails}) = \frac{22}{50} = 0.44 = 44\%$$

8. What is the experimental probability that the spinner lands on a primary color (red, blue, or yellow) and the coin lands tails up, $P(\text{primary color, tails})$?

Answers will vary.

$$P(\text{primary color, tails}) = \frac{16}{50} = 0.32 = 32\%$$

9. Explain how these experimental probabilities compare to the theoretical probabilities you calculated using the tree diagram.

Answers will vary. The experimental probability and theoretical probabilities slightly differ.

10. Why do you think the experimental probabilities differed from the theoretical probabilities?

Answers will vary. Theoretical probability is what we expect to happen. Experimental probability is what actually happens when we conduct the experiment.

14.4.2 EXPLORATION: CALCULATING THEORETICAL PROBABILITY (CONTINUED)

TEACHER MOVES

Compare and Connect

Consider prompting the students to work with another pair to compare probabilities. Prompt students to use mathematical language while discussing the following questions:

- What is the theoretical probability of each event?
- Are the experimental probabilities similar to each other?
- Are your experimental probabilities similar to the other group's?
- How do the experimental probabilities compare to the theoretical probabilities? Are there any that differ greatly?

ENRICHMENT

At the completion of the Exploration, consider using intentional questioning to solidify student understanding:

- If the theoretical probability of an event is $\frac{1}{2}$, but the experimental probability doesn't equal $\frac{1}{2}$, does that mean the theoretical probability should be changed?
- What is the difference between calculating theoretical and experimental probability?

14.4.3 COLLABORATION: COMPOUND PROBABILITY

TEACHER MOVES

Prompt students to use the tree diagram and previously calculated theoretical probabilities above to complete this table. Reinforce that this table is asking for theoretical probability.



14.4.3 Collaboration: Compound Probability

- With your partner, complete the probability table to show the theoretical probabilities of each step of an experiment and the overall probability of the event.

For example, in the P(yellow, heads) row, the theoretical probability of the first step of this event, spinning yellow on Spinner B, is $\frac{1}{5}$; therefore, $\frac{1}{5}$ is written in the "...First step" column. Next to that, the theoretical probability of landing on heads when a fair coin is flipped, $\frac{1}{2}$, is recorded. The overall theoretical probability, P(yellow, heads), $\frac{1}{10}$, is written in the "...Event" column.

Event		Theoretical Probabilities of the...		
		...First Step	...Second Step	...Event
Spin Spinner B, Flip a Coin	P(yellow, heads)	$\frac{1}{5}$	$\frac{1}{2}$	$\frac{1}{10}$
	P(not blue, tails)	$\frac{4}{5}$	$\frac{1}{2}$	$\frac{2}{5}$
	P(primary color, tails)	$\frac{3}{5}$	$\frac{1}{2}$	$\frac{3}{10}$

- Suppose a fair coin is tossed three times. Complete the table.

Event	Theoretical Probability of ...			
	First Toss is Heads	Second Toss is Heads	Third Toss is Heads	All Three Tosses are Heads
Flip three heads in a row	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{8}$

TEACHER MOVES

Notice and Wonder

After providing time for self-reflection on these "Notice and Wonder" questions, provide an opportunity for the students to collaborate by discussing their responses with their partner.

- What do you notice about the tables?
Answers will vary. You can multiply the probabilities of each step in the event to get the overall probability.
- What do you wonder about the tables?
Answers will vary. I wonder if this occurs every time.



14.4.4 Bring It Together: Calculating Theoretical Probability

- The theoretical probability of the outcomes in a multi-step experiment can be found by

○ adding

○ multiplying

 the

○ theoretical probability

○ experimental probability

 of the outcome for each step of the experiment.
- In a two-step experiment where a fair, six-sided die is rolled and a fair coin is flipped, the theoretical probability of rolling a 3 and the coin landing on tails, $P(3, \text{tail})$, is found by:

First, find the theoretical probability of getting a 3 on a fair

six-sided die, $P(3) =$

○ $\frac{1}{6}$

○ $\frac{1}{2}$

 . Next, find the

○ experimental

○ theoretical

 probability of a fair coin landing on

tails, $P(\text{tails}) =$

○ 0.500.

○ 0.250.

 Finally,

○ multiply

○ add

 the

probabilities of these steps to determine the probability of

the outcome, $\frac{1}{6} \cdot$

○ $\frac{1}{2}$

 $= \frac{1}{12}$.

So, $P(3, \text{tails}) = \frac{1}{12} =$

○ .083

 $= 8.3\%$.

14.4.4 BRING IT TOGETHER: CALCULATING THEORETICAL PROBABILITY

DIFFERENTIATE INSTRUCTION

This component is intended to be guided by the teacher. Use observational data from the 14.4.2 Exploration and 14.4.3 Collaboration to determine how much guidance is needed to address key ideas in this component.

ENRICHMENT

While completing this section, encourage synthesis through questioning:

- How does calculating theoretical probability compare to calculating experimental probability?
- How can you determine the theoretical probability of a multi-step experiment?

14.4.4 BRING IT TOGETHER: CALCULATING THEORETICAL PROBABILITY (CONTINUED)

TEACHER MOVES

Use this rule to check your answers above. Label events in P(3, tail) as 'A' and 'B.'

ENRICHMENT

As students complete this Bring It Together, prompt them with the following:

- What do you notice about these calculations? How are they similar? How are they different?
- What does 64 represent?
- How can you be certain that you have chosen the correct probability?

In a two-step experiment where the results in one step do not influence the results in the other step,
 $P(A \text{ and } B) = P(\text{event } A) \cdot P(\text{event } B)$.

3. In a repeated experiment where a fair coin is tossed six times, the theoretical probability of landing on tails every time is found by
☐ adding
☒ multiplying
 the theoretical probability of landing on tails for each trial.

$P(\text{all tails}) =$ _____

- $P\left(\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}\right) = \left(\frac{1}{2}\right)^6 = \frac{1}{64} = 0.015625 = 1.5625\%$.
- $2^6 = 64$.

**14.4.5 Wrap-Up: Cool Down**

1. Write a short note to a friend who missed today's class summarizing what you learned.

Answers will vary. We learned that in a repeated experiment in which the results in one step do not influence the results in the other step, we can multiply the probability of each step in order to get the overall probability.

14.4.5 WRAP-UP: COOL DOWN**DIFFERENTIATE INSTRUCTION**

Consider providing sentence stems for this prompt:

- Today we learned _____.
- We can calculate _____.
- In class, we completed _____ to learn _____.
- This can be applied to _____.